

QRAFT TECHNOLOGIES, LONG-TERM EQUITY TEAM, 2024

Multi-Agent Weekly Risk Report

Three LLM agents on LangGraph that turn the Japan / Taiwan risk-signal outputs into a structured, client-ready weekly report — automatically, every Monday.

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Why automate the weekly report

A risk signal is a number; the fund team and the client need the reasoning behind it, every week



Numbers need a narrative

Model probabilities and a hedge ratio do not explain themselves. PMs and clients ask what changed and why.



Analyst hours, every week

Writing the commentary by hand took a researcher's Monday morning — and slipped when markets were busy.



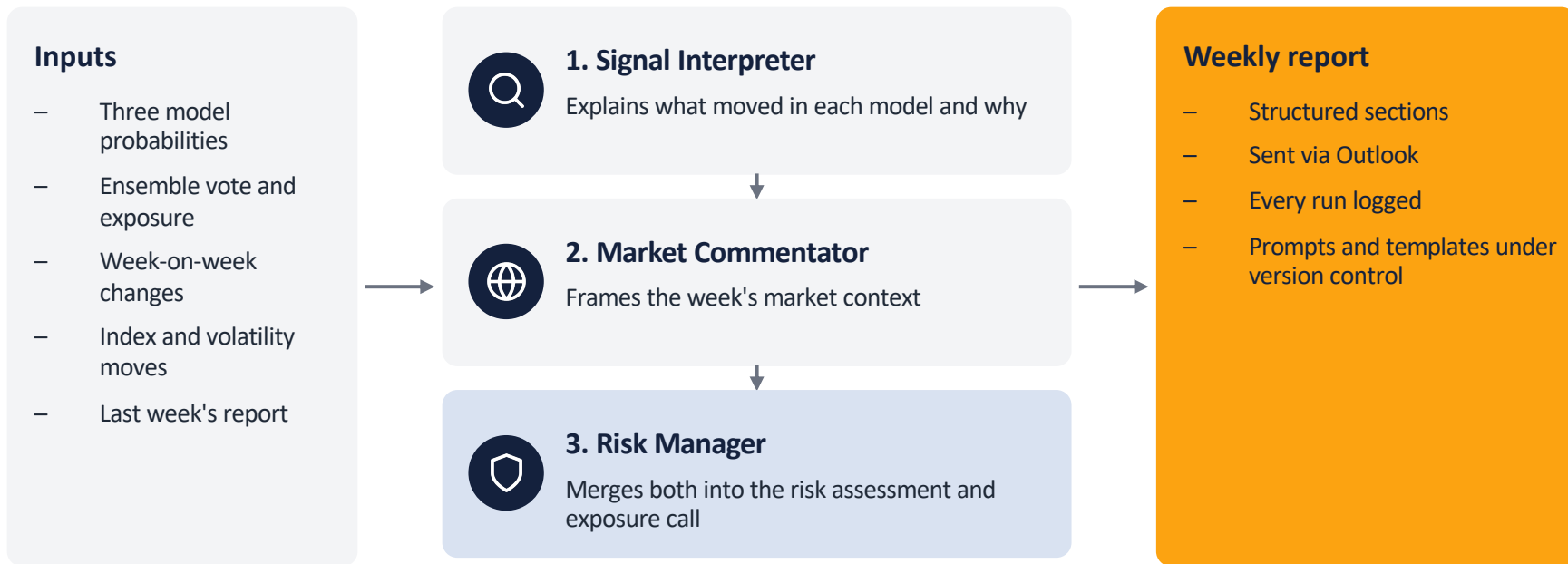
Consistency and audit trail

The same sections, the same logic, and a record of every input and output behind each report.

Goal: a report that reads as if written by the risk team, grounded in the signal tables, produced without manual work.

Architecture

Signal outputs flow through three specialised agents into one report



Orchestrated with LangGraph and GPT on Qraft's in-house agent harness.

From the signal table to sentences

Illustrative example — the values are placeholders, not a real report

Signal	This week	Last week
Sentiment (Model 1)	0.71	0.62
Momentum (Model 2)	0.44	0.41
Turbulence (Model 3)	0.18	0.20
Ensemble (average)	0.44	0.41
Equity exposure	56%	59%



Signal Interpreter

"Sentiment risk rose to 0.71 (+0.09 week on week) as put open interest climbed; momentum (0.44) and turbulence (0.18) were little changed, so the ensemble moved to 0.44 and equity exposure to 56%."

Market Commentator

"Taiwan equities fell on the week, led by semiconductor names; index implied volatility rose while realised volatility stayed moderate."

Risk Manager

"Risk is rising but not broad-based. Trim exposure to 56% in line with the ensemble; revisit if turbulence confirms the move."

The three agents

Narrow roles, explicit inputs, and guardrails that keep the text tied to the numbers

Agent	Reads	Produces	Guardrails
Signal Interpreter	Model probabilities, ensemble vote, week-on-week deltas	Model-by-model explanation of what changed and the resulting exposure	Every statement cites a value from the signal table; no market opinions
Market Commentator	Index returns, realised and implied volatility, macro calendar	Concise context for the week in each market	Context only — no positioning or recommendations
Risk Manager	Both agents' outputs plus the raw signal	Risk assessment and recommended exposure, with the reasoning	Must agree with the numeric signal; flags any disagreement for review

Weekly cycle and operations

From Friday's data to Monday's report



Reproducibility

Every run stores inputs, prompts and outputs; prompt and template changes are versioned so any past report can be traced.



Human in the loop

The Risk Manager agent's disagreement flag routes the report to a researcher before it goes out; otherwise delivery is automatic.

Report structure

The same five sections every week, so readers know where to look

Weekly Risk Report — Japan & Taiwan

1 Signal summary — this week's call and exposure

2 Model-by-model changes

3 Market context

4 Risk assessment and recommended exposure

5 Appendix — signal tables

- Sections 1–2 are written by the Signal Interpreter, 3 by the Market Commentator, 4 by the Risk Manager
- Every number in the text is copied from the signal tables in the appendix, never re-computed by the model
- Tone and length are fixed by the template so reports are comparable week to week
- The Outlook message carries the report and the run identifier for traceability

Impact

The weekly FSITC risk commentary moved from a manual write-up to an automated, reviewed pipeline.

Explanations are consistent and grounded: each agent has one job and cites the signal it reads.

Logging and version control make every past report reproducible.

The same pattern — specialist agents behind a supervisor — was reused later for enterprise multi-agent systems.

LangGraph, GPT, Python, Qraft in-house agent harness

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